

Linear Regression Models Recitation Week 3

Question 1

The mean square error of simple linear regression, denoted by MSE, is defined as

$$\text{MSE} = \frac{\text{SSE}}{n-2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}.$$

Show that MSE is an unbiased estimator of σ^2 ; that is, show that

$$\mathbb{E}(\text{MSE}) = \sigma^2.$$

Solution 1

Proof. Consider the simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where $x_1, \dots, x_n$ are fixed and

$$\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i = 1, \dots, n.$$

Define

$$\bar{y} := \frac{1}{n} \sum_{i=1}^n y_i, \quad \bar{x} := \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{\varepsilon} := \frac{1}{n} \sum_{i=1}^n \varepsilon_i.$$

Averaging the regression model over i gives

$$\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{\varepsilon}.$$

Also, let

$$S_{xx} := \sum_{i=1}^n (x_i - \bar{x})^2,$$

and assume $S_{xx} > 0$. Substituting the model for y_i and the expression above for $\bar{y}$ into the ordinary least-squares slope estimator gives

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x}) [(\beta_0 + \beta_1 x_i + \varepsilon_i) - (\beta_0 + \beta_1 \bar{x} + \bar{\varepsilon})]}{S_{xx}}. \end{aligned}$$

The intercept terms cancel inside the brackets, so

$$\begin{aligned}\widehat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x}) [\beta_1(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon})]}{S_{xx}} \\ &= \frac{\beta_1 \sum_{i=1}^n (x_i - \bar{x})^2}{S_{xx}} + \frac{\sum_{i=1}^n (x_i - \bar{x}) \varepsilon_i}{S_{xx}} - \frac{\bar{\varepsilon} \sum_{i=1}^n (x_i - \bar{x})}{S_{xx}}.\end{aligned}$$

By the definitions of S_{xx} and $\bar{x}$,

$$\sum_{i=1}^n (x_i - \bar{x})^2 = S_{xx} \quad \text{and} \quad \sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Therefore,

$$\widehat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) \varepsilon_i}{S_{xx}}.$$

Define

$$\delta := \widehat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) \varepsilon_i}{S_{xx}}.$$

Since

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x},$$

we have

$$\widehat{\beta}_0 - \beta_0 = \bar{\varepsilon} - \bar{x} \delta.$$

Therefore, the i th residual is

$$\begin{aligned}y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i &= \varepsilon_i - (\widehat{\beta}_0 - \beta_0) - (\widehat{\beta}_1 - \beta_1) x_i \\ &= \varepsilon_i - \bar{\varepsilon} - (x_i - \bar{x}) \delta.\end{aligned}$$

It follows that

$$\begin{aligned}\text{SSE} &= \sum_{i=1}^n \{\varepsilon_i - \bar{\varepsilon} - (x_i - \bar{x}) \delta\}^2 \\ &= \sum_{i=1}^n (\varepsilon_i - \bar{\varepsilon})^2 - 2\delta \sum_{i=1}^n (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon}) + \delta^2 S_{xx}.\end{aligned}$$

Since

$$\sum_{i=1}^n (x_i - \bar{x}) = 0,$$

we have

$$\begin{aligned}\sum_{i=1}^n (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon}) &= \sum_{i=1}^n (x_i - \bar{x}) \varepsilon_i \\ &= \delta S_{xx}.\end{aligned}$$

Thus,

$$\text{SSE} = \sum_{i=1}^n (\varepsilon_i - \bar{\varepsilon})^2 - \delta^2 S_{xx}.$$

We now take expectations. First,

$$\sum_{i=1}^n (\varepsilon_i - \bar{\varepsilon})^2 = \sum_{i=1}^n \varepsilon_i^2 - n\bar{\varepsilon}^2.$$

Since $\mathbb{E}(\varepsilon_i^2) = \sigma^2$ and $\text{Var}(\bar{\varepsilon}) = \sigma^2/n$, we obtain

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^n (\varepsilon_i - \bar{\varepsilon})^2 \right] &= n\sigma^2 - n\frac{\sigma^2}{n} \\ &= (n-1)\sigma^2. \end{aligned}$$

Next, $\mathbb{E}(\delta) = 0$, and

$$\begin{aligned} \text{Var}(\delta) &= \frac{1}{S_{xx}^2} \text{Var} \left(\sum_{i=1}^n (x_i - \bar{x}) \varepsilon_i \right) \\ &= \frac{\sigma^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{\sigma^2}{S_{xx}}. \end{aligned}$$

Hence,

$$\mathbb{E}(\delta^2 S_{xx}) = S_{xx} \text{Var}(\delta) = \sigma^2.$$

Consequently,

$$\mathbb{E}(\text{SSE}) = (n-1)\sigma^2 - \sigma^2 = (n-2)\sigma^2.$$

Finally,

$$\mathbb{E}(\text{MSE}) = \frac{1}{n-2} \mathbb{E}(\text{SSE}) = \sigma^2.$$

Therefore, MSE is an unbiased estimator of σ^2 . □

Question 2

Recall that the sample residual is defined by

$$e_i = y_i - \hat{y}_i,$$

where y_i is the i th response value and

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

is its corresponding fitted value computed using the least-squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$. Prove the following properties:

(i)

$$\sum_{i=1}^n x_i e_i = 0.$$

(ii)

$$\sum_{i=1}^n \hat{y}_i e_i = 0.$$

Solution 2

Part (i)

Proof. The least-squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ minimize the residual sum of squares

$$Q(b_0, b_1) := \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

over $b_0, b_1 \in \mathbb{R}$. Because Q is differentiable on $\mathbb{R}^2$, its gradient must vanish at a least-squares minimizer. Differentiating with respect to the intercept gives

$$\begin{aligned} 0 &= \left. \frac{\partial Q}{\partial b_0} \right|_{(b_0, b_1) = (\hat{\beta}_0, \hat{\beta}_1)} \\ &= -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \\ &= -2 \sum_{i=1}^n e_i. \end{aligned}$$

Hence the residuals sum to zero:

$$\sum_{i=1}^n e_i = 0.$$

Similarly, differentiating with respect to the slope gives

$$\begin{aligned} 0 &= \left. \frac{\partial Q}{\partial b_1} \right|_{(b_0, b_1) = (\hat{\beta}_0, \hat{\beta}_1)} \\ &= -2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \\ &= -2 \sum_{i=1}^n x_i e_i. \end{aligned}$$

Therefore,

$$\sum_{i=1}^n x_i e_i = 0,$$

which proves part (i). $\square$

Part (ii)

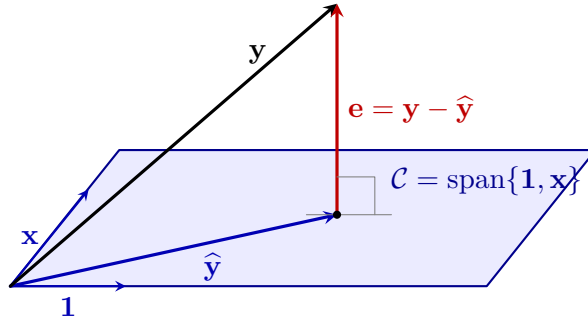
Proof. Use the fitted-value identity $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the two normal equations established in Part (i). Then

$$\begin{aligned} \sum_{i=1}^n \hat{y}_i e_i &= \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i) e_i \\ &= \hat{\beta}_0 \sum_{i=1}^n e_i + \hat{\beta}_1 \sum_{i=1}^n x_i e_i \\ &= \hat{\beta}_0 \cdot 0 + \hat{\beta}_1 \cdot 0 \\ &= 0. \end{aligned}$$

Thus the residual vector is orthogonal to the fitted-value vector, which proves part (ii). $\square$

Geometric intuition

With an intercept, the regression space is $\mathcal{C} = \text{span}\{\mathbf{1}, \mathbf{x}\}$. Least squares projects $\mathbf{y}$ orthogonally onto this space.



Because $\mathbf{e} \perp \mathcal{C}$, while $\mathbf{x}, \hat{\mathbf{y}} \in \mathcal{C}$,

$$\mathbf{x}^T \mathbf{e} = 0, \quad \hat{\mathbf{y}}^T \mathbf{e} = 0;$$

equivalently,

$$\sum_{i=1}^n x_i e_i = 0, \quad \sum_{i=1}^n \hat{y}_i e_i = 0.$$